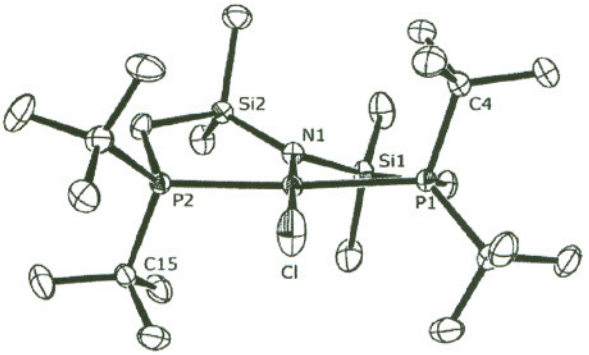
# Inorganic Cume - 11/7/03




1) Caulton and co-workers recently reported the crystal structure of the ruthenium phosphine complex (**1**) shown to the right (*JACS* **2003**, 8426). Except for the Ru, all unlabeled atoms are carbons. What is the oxidation state of the ruthenium?

2) Provide the electron counting for this metal complex.

3) This ($PNP^{tBu}$)RuCl complex (**1**) was found to be paramagnetic. There was some concern that the compound in question actually contained a hydride ligand and that the correct structure was in fact ($PNP^{tBu}$)RuHCl (**2**). Give at least two reasons why, based on the information provided, this was a concern. Name three different methods that can be used to distinguish between **1** and **2**.

4) Dissolution of **1** in the strongly coordinating PhCN solvent leads to a diamagnetic complex. Propose a structure for this new compound.

5) A similar phosphine ligand, $PNP^{Cy}$, has been used to prepare the compound ($PNP^{Cy}$)RuH($H_2$). Draw the structure of this molecule and provide the oxidation state of the metal.